

Supply Chain & Disruptions



Battery Electronic Vehicles



Customer/ Supplier misalignment



Long Development times



Supplier delays or unpreparedness



Logistics impact

Solutions

Supplier collaboration platforms, Reduce dependency on single suppliers by sourcing from multiple regions.

Real-time supply chain visibility using IoT and AI. Industry 4.0

Strengthening collaborations with Tier 1 & Tier 2 suppliers for resilience.

Nearshoring & Localization: Moving production closer to demand centres to reduce logistics risks.

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information about the car, but later it started alerting me to my driving style. On some days, it rated my behaviour as poor or moderate. When I eventually asked what the system was tracking, I was told it monitored mileage trends and even key driving patterns. It was a reminder that our cars now monitor us almost all the time, often more than we realise.

This shift to software-defined vehicles is driving similar changes in safety systems, artificial intelligence, and connected technologies. These advances are placing increasing pressure on engineering and manufacturing teams to work faster and more accurately.

Where technology meets process

This is where NPI comes in. Even the most innovative product has little value unless it can be manufactured at scale. NPI is where technology meets process. The industry talks extensively about ideation, concept development, and validation. However, the real work begins when a promising design has to move into production.

The industry faces several challenges. Regulatory pressure is intense, particularly with the rapid transition towards sustainability. Many in India will remember how

disruptive the transition to BS6 was. The automotive industry experienced a temporary slowdown because the shift required significant technological changes within a very short timeframe.

Supply chain disruption has been a persistent challenge. Before the Covid-19 pandemic, supplier delays were frustrating but generally manageable. Suppliers could often be persuaded to expedite deliveries. However, Covid-19 extended semiconductor lead times to 52, 80, or even 100 weeks. We had to rethink everything we knew about procurement.

Customer expectations have become another source of complexity. Customers now expect integrated functionality, advanced driver-assistance systems, highly customised experiences, and continuous feature updates. Even battery chemistry is evolving rapidly.

Engineering in a faster world

Late design changes, misalignment between digital designs and manufacturing requirements, high prototyping costs, and siloed systems create a challenging environment for modern engineering teams. This is where digital twins, virtual validation, concurrent engineering, and predictive analytics play a critical role.

Digital twins allow us to design both the product and the manufacturing process simultaneously. Cross-domain engineering ensures that software, electronics, mechanics, and materials evolve together. Simulations help us identify problems long before we build a physical prototype.

When these practices are fully integrated, they reduce risk, cost, and time.

The real challenge lies in achieving that integration.

Manufacturing must evolve at the same pace

Once the design has been frozen, manufacturing teams have limited scope to change production schedules. Production planning often begins before engineering is fully complete. Delays in design inevitably delay product launches, making early-stage design decisions especially critical.

Robotics, artificial intelligence, and connected systems are enabling smart factories and helping ease this pressure. However, they also create new skills gaps. Mechanical expertise alone is no longer enough. Today, we need technicians with expertise in software, analytics, and digital workflows.

Another area that remains a weakness for many companies is engineering change management. A single missed document or design update can lead to costly production errors.

Aligning design intent with manufacturing drawings and production processes remains a significant industry-wide challenge.

The supply chain is now a strategic asset

The supply chain has evolved from